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Weekly Report 1 due January 12, 2016

Hello! This was the first week of the laboratory; as such I spent most of the time lost in the “newness” of everything. This being the first robot of any sort that I have put together, I initially was a little overwhelmed and spent most of this week reading the textbook, reading/watching final presentations of past projects, talking to alumni of the class, and researching my options online. I realize that I'm a little bit behind schedule at this point, but I'll do my best to catch up this upcoming week.

This weekend, I spent about 4 hours assessing my options and trying to “finalize” the very abstract concept of my robot (I say “finalize” knowing that I may yet change what I do depending on the advice of the TA s/Professors): I'm currently thinking of building a robot pair that “plays catch” by throwing a small ball between each other. The idea is that the robots would find each other via a camera and computer vision, adjust for any obstacles in the path between them, and one would shoot a ball that the other would catch. In a perfect world, these robots would robustly respond to outside environments – by moving via wheels and adjusting the speed/angle of the launching mechanism – and transfer the ball object to the target robot. Ideally it would do so fast enough to respond to, say, a human standing in-between them trying to get the ball, similar to the child's “monkey in the middle” game.

I had wanted to try to sketch something out with SolidWorks, but after spending an hour with the program, I quickly realized that I would have to allot time to learn how the program worked more fully before attempting to seriously attempt a design. Nonetheless, I remain confident that sketching out the robot's design in detail is an important prerequisite to identifying and ordering the parts needed to construct the intelligent machine. Unfortunately, this also means that I have come to no hard decisions on things to purchase. What I do have in hand is a camera and an NVIDIA Jetson TK1 development board, which I will use to prototype the OpenCV needed to track the ball object/identify a partner robot.

All in all, I acknowledge that I am a bit behind the curve in preparation at this point, but I have great enthusiasm for this laboratory and have the utmost confidence that I can work my way back to where I need to be. Here's hoping for a great semester!